

Dynamics from Probability: Predictive Operators and the Koopman Picture

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Abstract

The classical approach to dynamics, following Koopman, describes how observable functions evolve under a known transformation T — but this presupposes access to T . The prior question is: given a current observation Q and a future observation F , what does an observer predict about F given Q , without assuming the dynamics?

Making prediction precise forces an answer. The canonical answer is the regular conditional distribution of F given Q , a Markov kernel $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$ realising $P(F \in \cdot \mid Q = q)$ across all outcomes simultaneously. Its existence requires disintegrating a joint distribution — which requires a σ -additive probability measure on the observable algebra. That measure is not assumed here; it is the object constructed in [2] from coherent structured observations satisfying collective exhaustion.

Once Π_Q exists, composing with Q gives the map $Q_* : \Omega \rightarrow \mathcal{P}(\beta)$ sending each sample point to its conditional law. The central result — the predictive factorization theorem — shows that Q_* is the minimal sufficient statistic for future prediction. Indexing over time, the kernels $\{\Pi_t\}$ compose; their composition rule is the Chapman–Kolmogorov equation, derived from temporal coherence rather than assumed. From it emerge a semigroup of Markov operators $\{K_t\}$ and the duality $\int K_t g d\mu = \int g d(\mu \star \Pi_t)$. When the kernels are Dirac measures, the construction recovers the classical Koopman operator exactly.

The output is a triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\})$ — predictive state space, stationary distribution, Markov semigroup. Setting $Q_t = h \circ T^t$ in a measure-preserving system identifies Q_* with the delay map $x \mapsto (h(T^n x))_{n \in \mathbb{Z}}$, making this triple the direct input to the reconstruction question studied in [3].

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1 Introduction

The classical approach to dynamics, following Koopman [1], describes how observable functions evolve: the operator $U_T f = f \circ T$ acts by precomposing with a known transformation T . This presupposes access to T as a primitive. The question this paper asks is prior: given a current observation $Q : \Omega \rightarrow \alpha$ and a future observation $F : \Omega \rightarrow \beta$, what does an observer predict about F given Q ? The dynamics, if there are any, should follow from the answer — not precede it.

Making prediction precise immediately generates a requirement. The canonical answer is the regular conditional distribution of F given Q : the unique Markov kernel $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$ realising $P(F \in \cdot \mid Q = q)$ simultaneously across all outcomes. Its existence rests on the Rokhlin disintegration theorem, which requires a σ -additive probability measure on the observable σ -algebra. That measure is not an extra assumption here. It is what [2] constructs: a coherent family of finitely additive charges on a directed system of observable algebras extends to a genuine probability measure exactly when it satisfies collective exhaustion, the condition ruling

out persistent mass on globally vanishing events. The predictive construction cannot begin until that measure exists; collective exhaustion is the condition under which it does.

Once Π_Q exists, composing with Q gives the map $Q_* := \Pi_Q(\cdot, -) \circ Q : \Omega \rightarrow \mathcal{P}(\beta)$, sending each sample point to its conditional law. The predictive factorization theorem shows that every conditional expectation of a function of F factors through Q_* : it is the minimal sufficient statistic for future prediction. Indexing over time, the kernels $\{\Pi_t\}$ compose; their composition rule is the Chapman–Kolmogorov equation, derived from temporal coherence rather than assumed. This yields a semigroup of Markov operators $\{K_t\}$ and the Koopman–Perron duality connecting them with the pushforward operators $\{\mu \mapsto \mu \star \Pi_t\}$. When the kernels are Dirac measures, the construction collapses to the classical Koopman picture.

The output is a triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\})$. Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ identifies Q_* with the delay map $\Phi_h(x) = (h(T^n x))_{n \in \mathbb{Z}}$, and this triple becomes the starting point for the reconstruction question of [3]: when does the delay orbit of a single observable determine the full state space?

2 Setup

The paper is self-contained. The connection to [2] is motivational; the formalism starts from a probability space $(\Omega, \mathcal{F}, P)$ and two measurable maps:

- $Q : \Omega \rightarrow \alpha$, where $(\alpha, \mathcal{A})$ is a measurable space;
- $F : \Omega \rightarrow \beta$, where $(\beta, \mathcal{B})$ is a standard Borel space.

The standard Borel hypothesis on β is used once: it is the hypothesis under which the Rokhlin disintegration theorem [4] guarantees the existence of a regular conditional distribution. All other results hold in greater generality.

We write $\mathcal{P}(\beta)$ for the space of probability measures on β , equipped with its canonical (Giry) σ -algebra. The joint pushforward $\rho := P \circ (Q, F)^{-1}$ on $\alpha \times \beta$ has marginals $\rho_\alpha = P \circ Q^{-1}$ and $\rho_\beta = P \circ F^{-1}$.

The main theorems are formalized in Lean 4 / Mathlib [5] in `PredictiveState.lean` and `PredictiveOperators.lean`. Several further results — the experiment preorder, general predictive sufficiency, the continuous-time generator — are stated but not yet formalized; these are noted at the relevant points.

3 The predictive kernel and the minimal state map

The regular conditional distribution of F given Q is the unique (up to $P \circ Q^{-1}$ -null sets) Markov kernel $\Pi_Q : \alpha \rightarrow \mathcal{P}(\beta)$ satisfying

$$\int_A \Pi_Q(q, B) d(P \circ Q^{-1})(q) = P(Q \in A, F \in B)$$

for all measurable $A \subseteq \alpha$ and $B \subseteq \beta$.

Definition 3.1 (Predictive kernel). The *predictive kernel* is Π_Q , the disintegration kernel of $\rho = P \circ (Q, F)^{-1}$ over its first marginal: $\Pi_Q(q, A) = P(F \in A \mid Q = q)$.

Proposition 3.2 (Predictive compatibility). *Let $\pi : \alpha \rightarrow \alpha'$ be measurable with $Q' = \pi \circ Q$. Then for $(P \circ Q^{-1})$ -almost every $q \in \alpha$:*

$$\Pi_Q(q, \cdot) = \Pi_{Q'}(\pi(q), \cdot).$$

Proof. The joint pushforward $\rho' = P \circ (Q', F)^{-1}$ satisfies $\rho' = \rho \circ (\pi \times \text{id})^{-1}$. The claim follows from the $P \circ Q^{-1}$ -a.e. uniqueness of the disintegration of ρ' over its first marginal, combined with the covariance of the conditional kernel under measurable maps (`eq_condKernel_of_measure_eq_compProd` in Mathlib). $\square$

Composing Π_Q with Q gives the map sending each sample point to its conditional law.

Definition 3.3 (Minimal predictive state map). The *minimal predictive state map* is

$$Q_* : \Omega \rightarrow \mathcal{P}(\beta), \quad Q_*(\omega) = \Pi_Q(Q(\omega), \cdot).$$

Proposition 3.4 (Measurability). Q_* is measurable.

Proof. Π_Q is a Markov kernel, hence measurable (`Kernel.measurable`). The wrap as a `ProbabilityMeasure` is measurable by `Measurable.subtype_mk`; measurability of Q_* follows by composition. $\square$

4 Predictive factorization

Theorem 4.1 (Predictive factorization). For every bounded measurable $g : \beta \rightarrow \mathbb{R}$:

$$E[g(F) \mid \sigma(Q)] = \int g dQ_*(\cdot) \quad P\text{-a.e.}$$

Proof. Let $h(\omega) := \int g dQ_*(\omega)$. We verify the two defining properties.

Measurability. The map $q \mapsto \int g d\Pi_Q(q, \cdot)$ is strongly measurable (`StronglyMeasurable.integral_kernel`) since $Q_* = \Pi_Q(Q(\cdot), \cdot)$, the map h is $\sigma(Q)$ -measurable.

Set-integral identity. For $S = Q^{-1}(A)$:

$$\begin{aligned} \int_S h dP &= \int_A \int g d\Pi_Q(q, \cdot) d(P \circ Q^{-1})(q) \quad (\text{pushforward}) \\ &= \int_{A \times \beta} g(f) d\rho(q, f) \quad (\text{disintegration: } \text{Measure.setIntegral_condKernel}) \\ &= \int_S g(F) dP. \end{aligned}$$

The conclusion follows from a.e. uniqueness of conditional expectation (`ae_eq_condExp_of_forall_setIntegral`). $\square$

Corollary 4.2 (Predictive sufficiency). $E[g(F) \mid \sigma(Q)] = E[g(F) \mid \sigma(Q_*)]$ P -a.e.

Proof. $\sigma(Q_*) \subseteq \sigma(Q)$, and the conditional expectation given $\sigma(Q)$ is already $\sigma(Q_*)$ -measurable by the theorem. $\square$

The operator $(K_{Q_*}g)(q_*) = \int g dq_*$ on $\mathcal{P}(\beta)$ recasts the theorem as: $E[g(F) \mid \sigma(Q)] = (K_{Q_*}g) \circ Q_*$ P -a.e.

5 Semigroup and duality

To introduce dynamics, index over a discrete time horizon. For each $t \in \mathbb{N}$, let $\Pi_t : \gamma \rightarrow \mathcal{P}(\gamma)$ be a Markov kernel on the state space γ , with $\Pi_0(q, \cdot) = \delta_q$. A family $\{\Pi_t\}$ satisfying the Chapman–Kolmogorov equation

$$\Pi_{t+s}(q, \cdot) = \int_{\gamma} \Pi_t(q', \cdot) d\Pi_s(q, q') \quad (\text{i.e., } \Pi_{t+s} = \Pi_t \circ_{\kappa} \Pi_s)$$

is a *Markov semigroup kernel*. The associated *Markov operators* are

$$(K_t g)(q) := \int g d\Pi_t(q, \cdot), \quad g : \gamma \rightarrow \mathbb{R}.$$

Proposition 5.1 (Basic properties). *Each K_t is positive, unital, and contractive on bounded measurable functions, with $K_0 = \text{id}$.*

Proof. $K_0 g(q) = \int g d\delta_q = g(q)$. Positivity, unitality, and contractiveness follow immediately from the Markov property. $\square$

Theorem 5.2 (Semigroup property). *$K_{t+s}g = K_t(K_sg)$ for all bounded measurable g .*

Proof. $(K_{t+s}g)(q) = \int g d(\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \int (\int g d\Pi_t(q', \cdot)) d\Pi_s(q, q') = \int (K_tg)(q') d\Pi_s(q, q') = (K_t(K_sg))(q)$, where the exchange of integrals is `Kernel.integral_comp`. $\square$

The pushforward of a measure μ on γ through the kernel is $P_t^*\mu := \mu \star \Pi_t = \int \Pi_t(q, \cdot) d\mu(q)$.

Theorem 5.3 (Koopman–Perron duality). *For every bounded measurable g and finite measure μ :*

$$\int (K_tg) d\mu = \int g d(P_t^*\mu).$$

Proof. $\int g d(P_t^*\mu) = \int g d(\mu \star \Pi_t) = \int_\gamma \int g d\Pi_t(q, \cdot) d\mu(q) = \int (K_tg) d\mu$, by Bochner–Fubini (`Kernel.integral_comp`). $\square$

6 Deterministic limit and the observable dynamical system

When $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$ for a measurable map $\varphi_t : \gamma \rightarrow \gamma$, the stochastic picture collapses.

Theorem 6.1 (Deterministic limit). *If $\Pi_t(q, \cdot) = \delta_{\varphi_t(q)}$, then $K_tg = g \circ \varphi_t$ and $\varphi_{t+s} = \varphi_t \circ \varphi_s$.*

Proof. $(K_tg)(q) = \int g d\delta_{\varphi_t(q)} = g(\varphi_t(q))$ (`integral_dirac'`). The flow law follows from $\delta_{\varphi_{t+s}(q)} = (\Pi_t \circ_\kappa \Pi_s)(q, \cdot) = \delta_{\varphi_t(\varphi_s(q))}$, and injectivity of the Dirac embedding (`SeparatesPoints`). $\square$

In the deterministic case, $K_tg = g \circ \varphi_t = U_T^t g$: the Markov operator specialises to the classical Koopman operator. The Koopman–Perron duality $\int K_tg d\mu = \int g d(P_t^*\mu)$ becomes the change-of-variables formula $\int g \circ T^t d\mu = \int g d(T_*^t\mu)$.

The full structure produced by the construction is:

Definition 6.2 (Observable dynamical system). An *observable dynamical system* is a triple $(\mathcal{P}(\beta), \nu_{Q_*}, \{K_t\}_{t \in \mathbb{N}})$ where $\mathcal{P}(\beta)$ is the predictive state space (the image of Q_*), $\nu_{Q_*} = P \circ Q_*^{-1}$ is the pushforward of P along Q_* , and $\{K_t\}$ is the Markov semigroup on $\mathcal{P}(\beta)$.

7 Reconstruction

Setting $Q_t = h \circ T^t$ in a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $h \in L^\infty$ specialises the construction. Under this choice, $Q_*(\omega) = (h(T^n\omega))_{n \in \mathbb{Z}}$ is the delay map Φ_h , and the observable dynamical system is the triple $(\Phi_h(X), \mu \circ \Phi_h^{-1}, \{K_t\})$.

The companion paper [3] asks: when does the delay orbit of a single h determine the full state space? The answer is the reconstruction theorem: the observable σ -algebra $\mathcal{O}_h = \sigma(\{h \circ T^n\})$ equals $\mathcal{B}(X)$ modulo μ if and only if the algebra generated by $\{h \circ T^n\}$ is dense in $L^2(X, \mu)$. The proof turns on the density bridge: the L^2 -closure of a function algebra equals the L^2 -space of the σ -algebra it generates.

The Koopman operators $\{U_T^t\}$ on $L^2(X, \mu)$ relate to the Markov operators $\{K_t\}$ on $\Phi_h(X)$: when Φ_h is injective modulo μ -null sets, they act on the same information. The cyclic vector condition $\overline{\text{span}}\{U_T^n h : n \in \mathbb{Z}\} = L^2(X, \mu)$ is sufficient for reconstruction but strictly stronger — reconstruction requires only that the algebra generated by the delay orbit be L^2 -dense, not the linear span. They coincide only when U_T has simple spectrum.

When reconstruction holds, $\Phi_h(X)$ is the full state space X up to measure-theoretic identification. The compact Stone space of [2] — introduced as a technical device for σ -additivity — is, under the reconstruction condition, identified with X itself. The construction begins with a compact completion of the sample space and ends by recognising it as the state space.

References

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